BS4 Series B Supplement

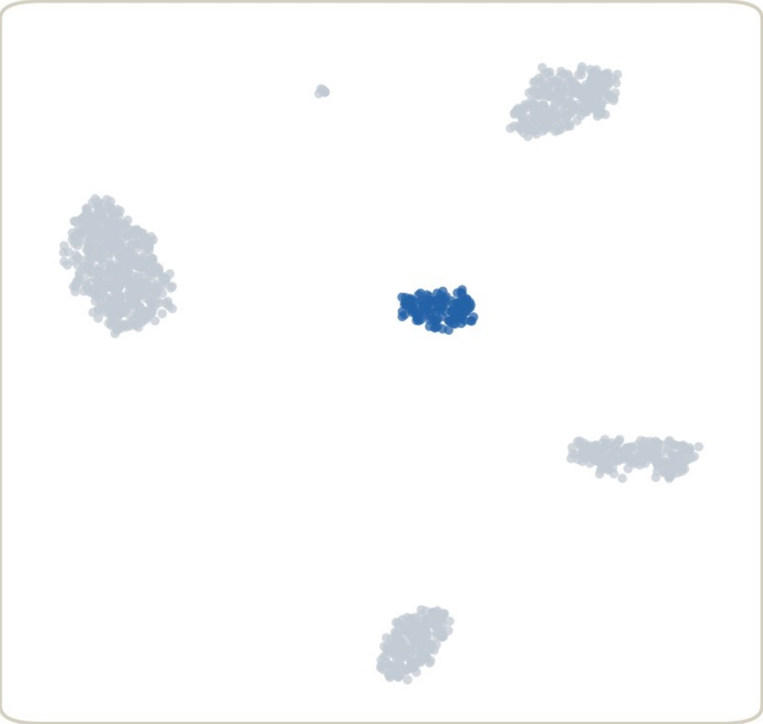
A field guide to annotation resources: from markers to atlases

Four resource classes, one decision tree, three field playbooks — annotation has no tool shortage; what you need is where to look, what to pick, and what to record.

About 35 min | Prerequisites: B10, B16 | Series B Supplement, Episode 4

One cluster, three names

your cluster 7 (CD3E+ / CD8A+)



blue = the cluster to name

tool 1: SingleR + HPCA

"NK cells"

tool 2: mapping (atlas reference)

"CD8 effector T"

tool 3: marker-score dictionary

"NKT-like cells"

which one is right?

all three are hypotheses, not answers

simulated data

one cluster, three resources, three names — the difference is not the tool, it is the reference behind each

All three resources are confident — the difference is not the tool, it is the reference behind each

WHERE THIS EPISODE STARTS

Three names.
Three hypotheses.

B10's rule has not changed: automated annotation is a hypothesis, not an answer. This episode is about where those hypotheses come from, how to verify them, and how to record them.

What you will learn

1. Sort the annotation-resource world into four classes — marker dictionaries, tools plus references, atlases, tumour-specific — and know each class's first pick and its traps

2. Walk the resource decision tree and assemble a resource kit for PBMC, tumour, and mouse data

3. Take away the 2026-09 maintenance snapshot and the four-things record discipline: tool, reference, version, date

01

Three iron rules of annotation

Before touring resources, put B10's and B16's rules on the table

The three rules

Automated labels are hypotheses

Every tool-issued label goes back to marker plots (DotPlot / FeaturePlot) for verification, and the four things get recorded (B10)

Malignancy rests on CNV

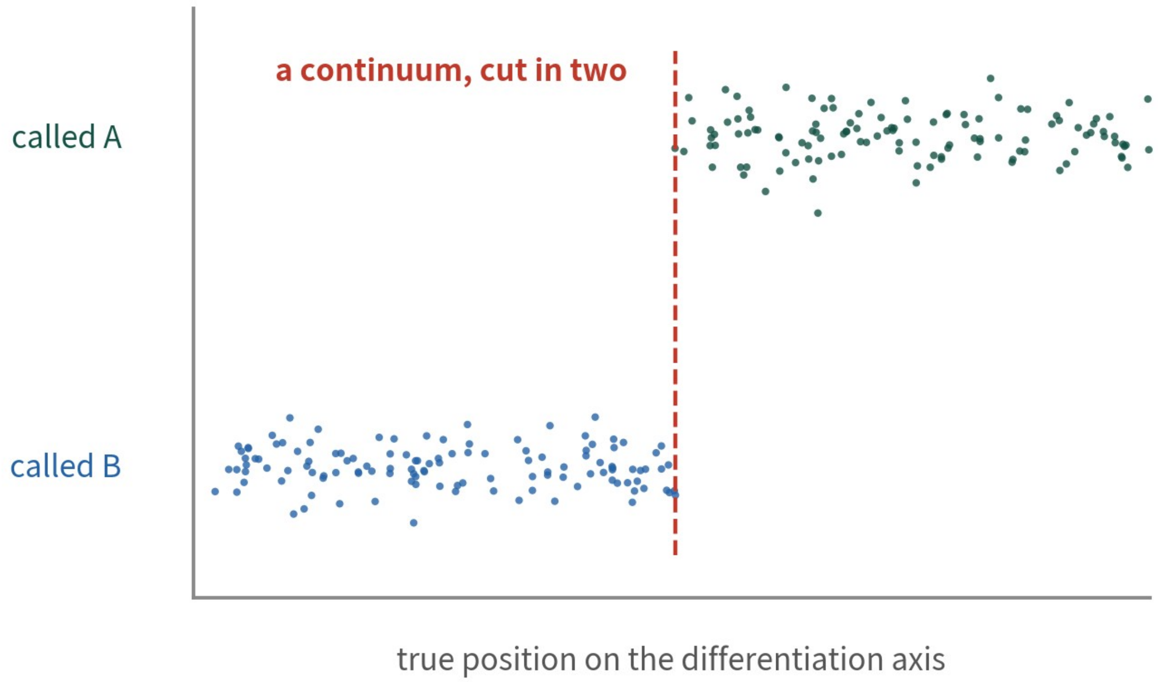
Tumour samples start with CNV inference plus cross-patient clustering behaviour; markers only name the microenvironment (B16)

A reference's nature sets its limits

Bulk-purified populations: solid on major classes, weak on continua; atlas mapping only knows the types the reference contains — disease states get forced into healthy labels

Rule three, demonstrated: bulk vs single-cell references

bulk reference (two purified means): a hard split



single-cell reference (sampled along the axis): the continuum survives

simulated data (correlation matching, actually run)

bulk references are solid on major classes, weak on continua and rare types — not a bug, the nature of purified populations

One differentiation continuum: the bulk reference cuts it in two, the single-cell reference preserves it — simulated data, correlation matching actually run

CORE IDEA

A reference's nature sets its limits.

The first question is never 'which is newest' — it is 'how was this reference built, and does it contain my types'.

02

Four classes, one map

The panorama first, then class by class — first stop: marker dictionaries

The annotation-resource quadrants

schematic

tumour context

manual look-up

automated matching

general tissue

IV. tumour-specific

inferCNV / CopyKAT / scATOMIC

3CA meta-programs • TISCH2

UCell / AUCell scoring

call malignancy first, then types

(the automated end of tumours)

scATOMIC: automated pan-cancer

scorers give relative enrichment only

malignancy still rests on CNV (B16)

automation exempts nothing

I. marker dictionaries

CellMarker (literature-level)

PanglaoDB (light, frozen)

the arsenal for manual checks

II. tools + refs / III. atlases

SingleR + celldex • CellTypist

Azimuth (15 refs) • scType

CELLxGENE • Tabula Sapiens • HLCA

generators of first opinions

each class does one job — day-to-day annotation is always an automated first opinion plus a manual marker check

Day-to-day annotation is always a combination: an automated first opinion plus a manual marker check

The four classes at a glance

Class	Representatives	When to use
I. Marker dictionaries	CellMarker, PanglaoDB	Manual cluster verification; literature provenance
II. Tools + reference sets	SingleR + celldex, CellTypist, Azimuth, scType	The automated first opinion, per cluster or per cell
III. Reference atlases	CELLxGENE, Tabula Sapiens, HLCA, Allen Brain	Find a reference for any tissue; subset and build your own
IV. Tumour-specific	inferCNV, CopyKAT, scATOMIC, 3CA, TISCH2, UCell/AUCell	Call malignancy first, then states; markers only name the TME

The two marker dictionaries

One for provenance, one for speed — different jobs, not either-or

CellMarker

The literature-level dictionary

- From the 2.0 paper: 26,915 markers, 2,578 cell types, 656 tissues
- 83,361 tissue-type-marker entries from 24,591 papers; 52,987 human, 32,285 mouse
- Every marker traces to its source paper — first stop when writing up
- Main site now 3.0; free, no login

PanglaoDB

The one-TSV lightweight table

- 8,286 associations, 178 types, 29 tissues — one file, one download
- First pick for teaching demos and quick AddModuleScore runs
- Marker table frozen since 2020-03-27
- Never the sole source in formal work — pair with newer databases

Trap 1: the dictionary version trap

What it is

Citing CellMarker scale numbers without a version, or leaning on PanglaoDB alone in formal work as if it were current.

Why it happens

CellMarker's main site is now 3.0 and the 2.0 download page 404s — check the official page for 3.0's formats; PanglaoDB stopped in 2020, so anything established since is simply absent.

What it costs you

A reviewer opens the official page and your numbers don't match; or your 'novel type' is missing from the dictionary and gets mislabelled under an old name.

03

Automated annotation: tools + references

Four tools, one celldex table, and one limit of mapping

The four automated annotators

The mechanics live in B10 — here we only decide which to reach for

Tool	Stack	One-line principle	Reach for it when
SingleR + celldex	R	Correlate against bulk-purified references	The first opinion in an R pipeline
CellTypist	Python	Pre-trained logistic-regression models, 77 of them	Scanpy pipelines; start immune data on Immune_All_Low
Azimuth	Web / R	Map onto a curated atlas, with confidence scores	Your tissue is on the official list of 15 references
scType	R	Reference-free, cluster-level marker scoring	A fast second opinion to cross-check SingleR

Choosing a celldex reference

celldex references are bulk-purified populations — solid on major classes, weak on continua and rare types

your data	pick this reference	why
PBMC	MonacoImmuneData	the vignette's pick for a typical PBMC sample
human, first pass	HumanPrimaryCellAtlasData	broad coverage (microarray-era atlas)
coarse but stable	BlueprintEncodeData	coarser labels, fewer mistakes
CD4+ T subsets	DICE	deep CD4 subset resolution
marrow / progenitors	NovershternHematopoieticData	highest myeloid / progenitor resolution
mouse immune	ImmGenData	extremely fine subtypes (ImmGen)
mouse general	MouseRNAseqData	brain, blood, heart and more

source: celldex (Bioconductor); every label still goes back to markers

Each of the seven has a home turf — MonacoImmune for PBMC is the vignette's own recommendation

The celldex selection table

Your data	Reference	Why, in one line
PBMC	MonacoImmuneData	The vignette's pick for a typical PBMC sample
Human, first pass	HumanPrimaryCellAtlasData	Broad coverage (microarray-era atlas)
Coarse but stable	BlueprintEncodeData	Coarser labels, fewer mistakes
CD4+ T subsets	DICE	Deep CD4 subset resolution
Marrow / progenitors	NovershternHematopoieticData	Highest myeloid / progenitor resolution
Mouse immune	ImmGenData	Extremely fine subtypes (the ImmGen project)
Mouse general	MouseRNAseqData	Brain, blood, heart and more

The other players, and the second-opinion rule

CellTypist (Python's first pick)

77 pre-trained models at about 1 MB each; strongest on immune, also lung (incl. HLCA), gut, brain; train custom models on your own data

scType (the fast second opinion)

Reference-free, cluster-level scoring; used via source() from GitHub, last release 2022 — a draft label and cross-check, never the final answer

The second-opinion rule

Any label that matters gets two independent sources — one correlation-based plus one mapping or dictionary-based; disagreements go to markers for the verdict

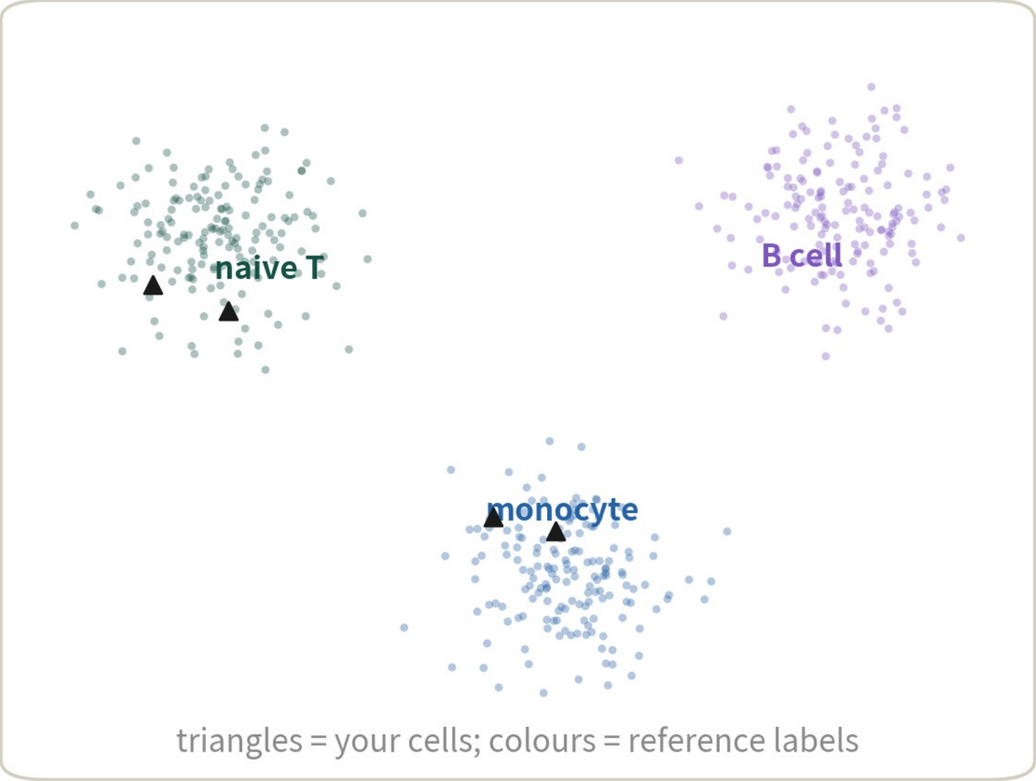
The R opener: SingleR meets celldex

```
library(SingleR); library(celldex)
ref <- MonacoImmuneData()     # PBMC 首選參考集
pred <- SingleR(
  test   = GetAssayData(obj, layer = "data"),
  ref    = ref,
  labels = ref$label.main,
  clusters = obj$seurat_clusters) # 第一輪用 cluster 模式
table(pred$labels)
```

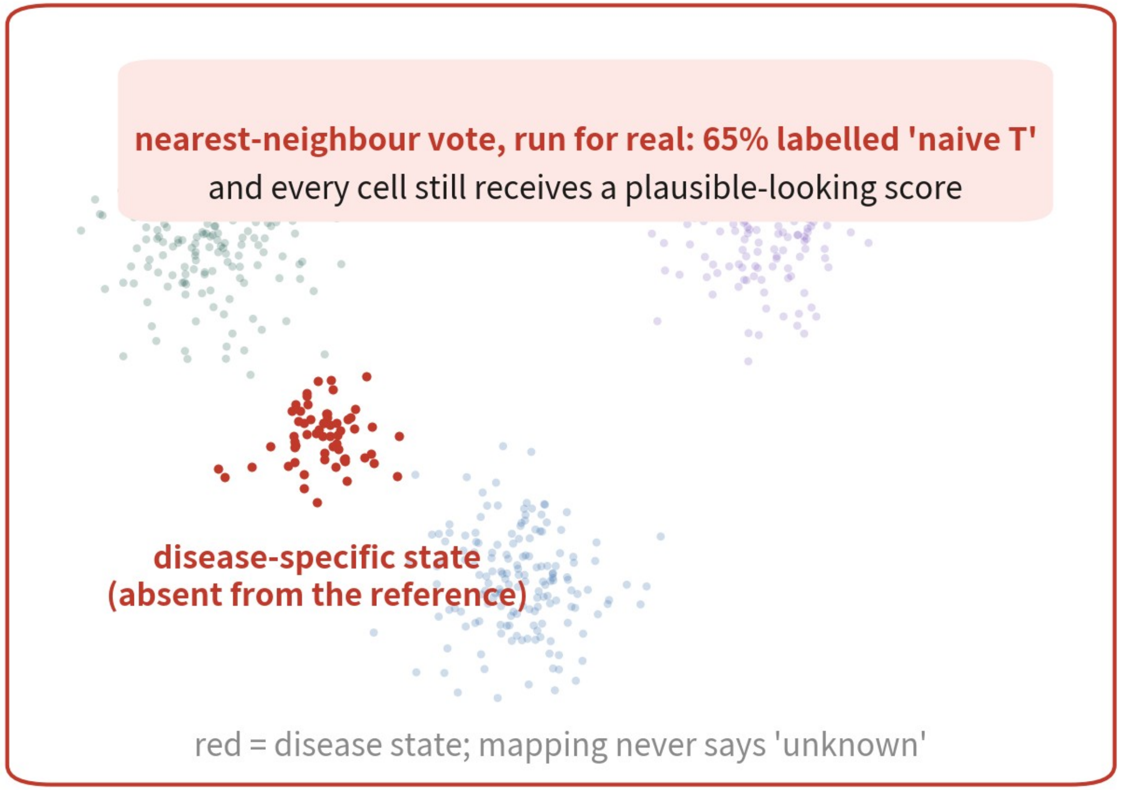
The full walkthrough — delta checks, Azimuth — is B10, slides 24–31; here we only confirm the opener

Mapping-style annotation: principle and limit

principle: project onto the reference, neighbours vote



the limit: types absent from the reference get forced in



simulated data

mapping-style annotation (Azimuth / HLCA): easiest inside the list — but it only knows what the reference contains

Azimuth ships 15 official references (PBMC: 161,764 cells with l1/l2/l3 labels) — but absent types get forced into present ones

Trap 2: treating mapping output as a census

What it is

Feeding disease samples to Azimuth / HLCA and accepting the healthy labels wholesale — without even glancing at the scores.

Why it happens

Mapping only knows what the reference contains and has no 'unknown' option: disease-specific states will be forced into the nearest healthy label, with plausible-looking scores.

What it costs you

Fibrosis-, peritumoural-, inflammation-specific states vanish from your results — the exact lesson Advanced 30 drills.

04

Atlases: where references come from

When your tissue is off every tool's list, the answer lives in an atlas portal

Four atlas entry points

CELLxGENE

The first stop for references

The homepage shows 33M+ unique cells, 436 datasets, 2.7K+ cell types; the Census snapshot adds 44M+ human cells; downloads are h5ad (rds being retired)

Tabula Sapiens

Pan-organ human

1.1M+ cells, 28 organs, 24 donors; v1 is the Science 2022 publication, v2 is still a preprint — say which one you cite

HLCA

The lung's atlas-grade reference

Core: 584,444 cells / 107 people / 14 datasets (Sikkema 2023); same source as Azimuth Lung v2; the h5ad is about 5.6 GB

Allen Brain

Brain-specific

Whole mouse-brain and human-brain atlases; public on AWS S3, no account, h5ad plus official Python tools — for when pan-organ atlases run out of resolution

From atlas to reference, in three steps

1 **1. Find the collection**

Search CELLxGENE by tissue; pick a dataset whose label depth fits your question — not the biggest, the best-labelled

2 **2. Download h5ad, convert**

rds / Seurat downloads are being retired; R users convert with zellkonverter — exactly what Basic 20 drills

3 **3. Subset and build the reference**

Subset the matching tissue as your reference, feed SingleR or Seurat TransferData — the method is B10/B11; this episode only supplies the material

05

Tumour-specific resources

Malignancy first, states second — never the other way round

The tumour trio: current status

the tumour trio: status as verified 2026-09

schematic

inferCNV	CopyKAT	scATOMIC
the classic CNV caller (B16's lead)	**automatic tumour / normal call**	**hierarchical pan-cancer labels**
officially unsupported; still installs from Bioconductor 3.21	README states: fails on paediatric and blood cancers (few CNAs)	trained on 19 cancers, validated on 13 (median F1 = 0.99)
formal work: add a cross-check	**fix: anchor with known normals**	**off-list cancers: no origin call**

the rule stands: malignancy = CNV + cross-patient clustering behaviour, never markers (B16)

the README names its successors: inferCNA / CopyKAT / Numbat (Numbat needs allele data — costlier)

near-diploid cancers like AML:

the CNV route often fails —
switch evidence (B19's AML playbook)

Each of the trio has a failure zone — status is the 2026-09 snapshot; reconfirm on the official pages before use

The trio, one by one

Tool	Role	Status and limits (2026-09)
inferCNV	The classic CNV caller (B16's lead)	README declares it unsupported and names inferCNA / CopyKAT / Numbat; Bioconductor 3.21 still ships v1.24.0 — fine for teaching, add a cross-check in formal work
CopyKAT	Bayesian automatic tumour/normal call plus subclones	README states it fails on paediatric and blood cancers (few CNAs); the fix is supplying known normal cells or anchoring on T cells
scATOMIC	Hierarchical pan-cancer labels plus malignancy calls	v2.0.3, maintained; trained on 19 cancers, externally validated on 13 (median F1 = 0.99); origin calls don't apply off-list

State annotation and look-up stations

3CA

The tumour-state headquarters

124 datasets, 40+ cancers, 5.54M cells, 149 meta-programs; the 41 malignant MPs (Neftel's GBM four states included) come in one XLSX — Gavish 2023, Table S2

TISCH2

The tumour-immune look-up

190 datasets, 6,297,320 cells, about 50 cancers; three annotation tiers (malignancy / major / minor); fast marker look-ups, but no batch download

UCell / AUCell

The scoring pair

Rank-based (UCell) and AUC-based (AUCell) signature scoring — pair with 3CA meta-programs or MSigDB C8; scores give relative enrichment only

Successors

Numbat / inferCNA

The README-named cross-check options; Numbat is haplotype-aware and needs allele data (BAMs), so it costs more to run

Trap 3: scores as malignancy calls

What it is

A cell scores high on a 'tumour signature' with UCell / AUCell, so it gets labelled malignant.

Why it happens

Scores are relative enrichment only — proliferation, stress, and metabolic signatures run high in normal cells too; the evidence for malignancy is CNV plus cross-patient clustering, never a signature.

What it costs you

Proliferating normal progenitors get called cancer, and your 'tumour heterogeneity' analysis runs on normal cells — the price of bypassing B16's rule.

06

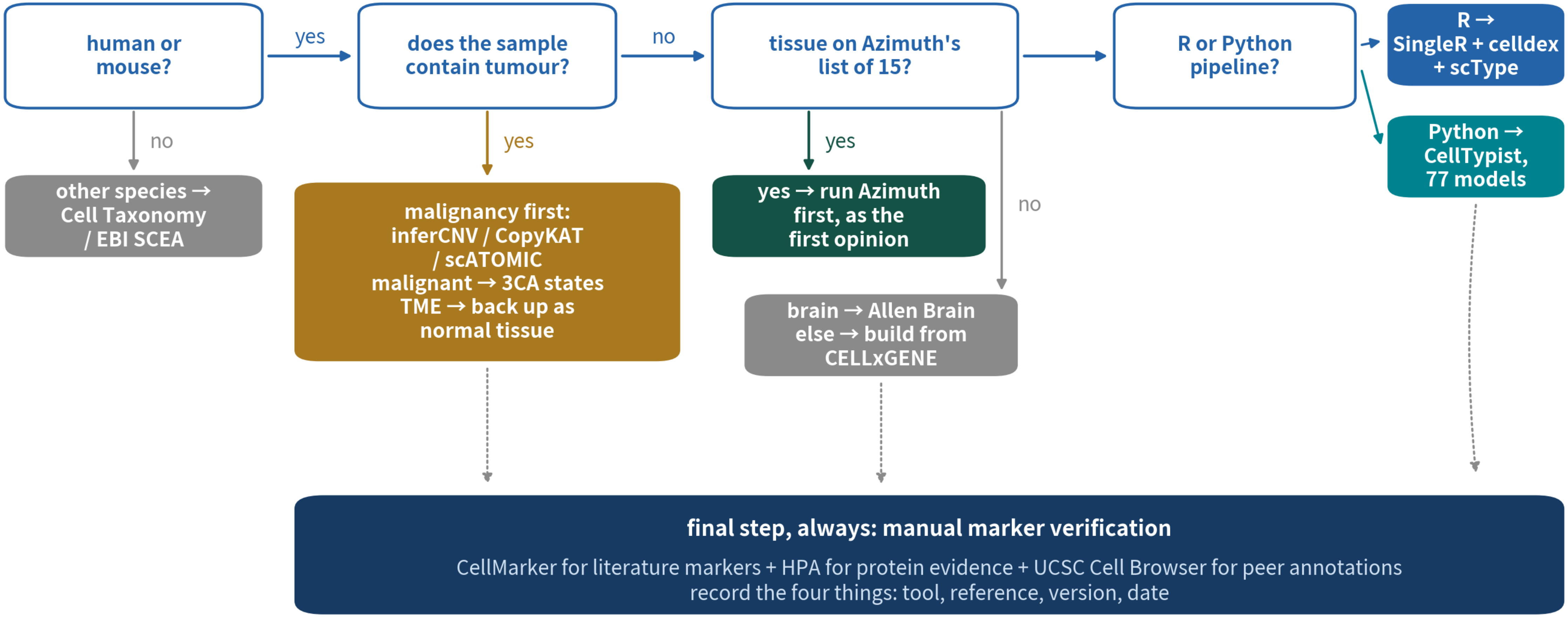
Walking the tree, three playbooks

Wiring the four classes into a path you can actually walk

The resource decision tree

the resource decision tree: start from your data

schematic



Four questions to the bottom, then markers, always — print it and keep it next to the library

The tree's five stations

1	**1. Species**	Human or mouse continue; other species go to Cell Taxonomy (34 species) or EBI SCEA (21 species), usually plus a hand-built marker list
2	**2. Tumour**	Tumour present → the trio calls malignancy first; malignant cells go to 3CA meta-programs, the TME continues as normal tissue
3	**3. Tissue**	On Azimuth's list of 15 → run it as the first opinion; brain → Allen Brain; anything else → build from CELLxGENE
4	**4. Ecosystem**	R → SingleR + celldex (pick the right reference) + scType as second opinion; Python → CellTypist (check the 77-model list first)
5	**5. Always**	Manual marker verification: CellMarker for provenance, HPA for protein evidence, UCSC Cell Browser for peer annotations; record the four things

Three field playbooks

three field playbooks

schematic

playbook 1: first PBMC annotation	Azimuth PBMC, web, first opinion	SingleR + MonacoImmune, second opinion	disagreeing clusters → CellMarker, manual verdict	record four things: tool • reference • version • date
playbook 2: a tumour sample	trio calls malignancy (CNV + clustering)	TME cells → run the normal-tissue route above	malignant cells → 3CA meta-programs + UCell scores	markers name the TME — never call malignancy
playbook 3: mouse data	immune-heavy → SingleR + ImmGen	general tissue → MouseRNAseqData	extra reference → Tabula Muris (20 organs)	CellMarker covers mouse markers too (32,285 entries)

all three playbooks end the same way: verify with markers, and record tool, reference, version, date

Three roads, one ending: verify with markers, record the four things

Playbook 1: your first PBMC annotation

First opinion: Azimuth

The web app needs no install; the PBMC reference has 161,764 cells with l1/l2/l3 labels — a draft in under half an hour

Second opinion: SingleR

Pick MonacoImmuneData from celldex; run cluster mode and cross-tabulate against Azimuth's labels

The verdict: CellMarker

Disagreeing clusters go back to canonical markers for the verdict (B10's judge routine); record the four things — drills: Basic 01 / 10 / 12

Playbook 2: a tumour sample

Malignancy first

Open with inferCNV, cross-check with CopyKAT / scATOMIC; the evidence is CNV plus cross-patient clustering (B16)

The TME goes the normal route

Immune and stromal cells run the normal-tissue route — Azimuth / SingleR plus marker checks, same as playbook one

Malignant cells get states

3CA meta-programs plus UCell scores name the states; markers never call malignancy — drills: Advanced 01 / 03 / 05 / 37

Playbook 3: mouse data

Immune-heavy

SingleR + ImmGenData for extremely fine subtypes; to chase one gene's expression across populations, go to the ImmGen site itself

General tissue

Open with MouseRNAseqData; top up with Tabula Muris when a reference is missing (20 organs, FACS and droplet)

The dictionary covers mouse

CellMarker holds 32,285 mouse entries — switching species exempts nothing from marker checks; drills: Basic 07 / 08 / 15 / 16 / 19

HOW ALL THREE END

The route may change.

Verification and records may not.

PBMC, tumour, and mouse walk three different roads — all ending at marker plots and a four-line record.

07

Maintenance snapshot, record discipline

Resources age — your records must not age with them

The maintenance snapshot (2026-09)

maintenance snapshot (annotation library, 2026-09)

inferCNV	**unsupported**	README says so; names inferCNA / CopyKAT / Numbat; Bioconductor 3.21 still ships v1.24.0
CellMarker	**main site now 3.0**	the 2.0 download page 404s; cite scale numbers as from the 2.0 paper (NAR 2023)
PanglaoDB	**effectively frozen**	marker table frozen at 2020-03-27 — fine for teaching, pair it with newer sources for real work
CELLxGENE	**rds being retired**	rds / Seurat downloads are being retired in favour of h5ad / Census — R users must convert h5ad
old URLs	**redirected**	portal.brain-map.org → brain-map.org; TISCH lives at tisch.compbio.cn; DISCO moved to the a-star domain

verified 2026-09-07. scales and URLs drift — spend 30 seconds on the official page before citing, and record the access date

All five lines come from the library's item-by-item verification on 2026-09-07 — this slide itself will expire; recheck before use

The four-things record card

the record card: four things

schematic

analysis notes • annotation record (mock)

tool	SingleR (cluster mode)
reference	celldex: MonacoImmuneData
version	paste the full sessionInfo()
date	accessed 2026-09-07

references get updated

CellMarker went 3.0, celldex turns over —
your record is the only way back

labels are hypotheses

when a reviewer asks where a label came from, these four lines are the answer

handovers must rerun

an unversioned annotation cannot be reproduced six months later, even by you

four things: tool, reference, version, date — written down for every annotation, same discipline as QC cutoffs

Tool, reference, version, date — the same discipline as QC cutoffs: numbers, reasons, records

Record discipline: three moves

1

1. Thirty seconds on the official page

Before citing any scale number or version, open the official page — this episode's snapshot is only guaranteed for 2026-09-07

2

2. Four things into the notes

Tool, reference, version, date; for versions, paste the full sessionInfo() — never from memory

3

3. Access dates into the report

Scale numbers cite their source version (CellMarker's numbers cite the 2.0 paper); download dates go into the methods section

Recap: the episode in one table

The job	First pick	The one line to remember
Verify markers by hand	CellMarker (+ HPA evidence)	PanglaoDB is frozen — light support only
Automated first opinion, R	SingleR + the right celldex reference	Bulk-purified nature: majors solid, continua weak
Automated first opinion, Python	CellTypist (start on Immune_All_Low)	77 models — check the list before you commit
Mapping, on-list tissues	Azimuth's 15 references	Absent types get forced into present labels
Tumour samples	The trio, then 3CA + UCell	Malignancy rests on CNV — scores don't count

Next steps

The annotation reference library

16 core + 20 extended entries + the xlsx quick table — the source of every number here; cross-species and protein-evidence scenarios live in the extended tier

Matching drills

PBMC annotation: Basic 01/10/12; mapping limits: Advanced 30; the tumour trio: Advanced 01/03/05/37; mouse: Basic 07/08 and friends

Method refreshers

How to run annotation is B10 (SingleR / Azimuth walkthrough); how to call malignancy is B16 (the full inferCNV run) — this episode does not reteach them

ONE-LINE SUMMARY

Tools turn over, atlases grow — the habit of verification never expires.

Four classes, one tree, a four-line record: carry these three and any resource turnover lands softly.

Check yourself 1

A tumour sample lands on your desk. Which resource class do you open first?

A Marker dictionaries — look up tumour markers

B Tools + references — run SingleR first

C Atlases — find a same-cancer atlas

D Tumour-specific — call malignancy first

Answer D. The order is fixed: call malignancy first with inferCNV / CopyKAT / scATOMIC plus cross-patient clustering; only then does the TME run the normal route and malignant cells get states. Markers never call malignancy — B16's rule.

Check yourself 2

On a hematopoietic continuum, SingleR + celldex forces transitional cells into the two endpoint types. The root cause?

A A bug in SingleR's algorithm

B The reference is bulk-purified population means — no intermediate states

C Insufficient sequencing depth

D Clustering resolution set too low

Answer B. celldex references are bulk-purified by nature: the reference holds only the mean profiles of A and B, so every transitional cell must be assigned to whichever end it resembles. A reference's nature sets its limits — a single-cell reference or atlas mapping preserves the continuum.

Check yourself 3

Six months on, a reviewer asks you to reproduce the annotation. What saves you?

A Remembering which computer you used

B The four things in your notes: tool, reference, version, date

C Rerunning the latest tool version

D Citing the reference library's URL

Answer B. References get updated — CellMarker went 3.0, celldex versions turn over, Azimuth references get revised. Rerunning the latest version may hand you different labels; only tool, reference, version, and date lead back to the original result.

NEXT EPISODE

BS5: A field guide to enrichment resources

One DE list, two websites, two completely different top pathways — who is lying? (Nobody.)

Gene-set databases, the KEGG licensing minefield, and one more decision tree. See you there.